Ex 6: Programs in Prolog

1. Family Relationships

Problem: Define family relations and answer grandparent queries.

```
% Facts
parent(john, mary).
parent(mary, susan).
parent(john, mike).
% Rules
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
% Query
% ?- grandparent(john, susan).
```

Output:
true

2: Animal Classification
Problem: Identify if an animal is a mammal based on features.

```
% Facts
has_hair(tiger).
gives_milk(tiger).
% Rule
mammal(X) :- has_hair(X), gives_milk(X).
% Query
% ?- mammal(tiger).
```

Output:
true.

3: Simple Arithmetic Knowledge Base
Problem: Implement simple addition using PROLOG.

```
% Rule
add(X, Y, Z) :- Z is X + Y.
% Query
% ?- add(5, 7, Z).
```

Output:
Z = 12.